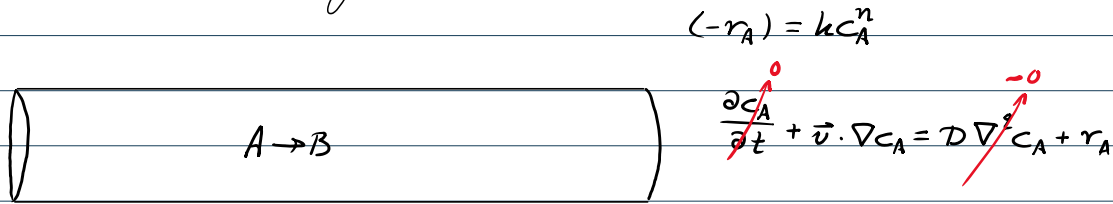


PFR simulation using simple PINN



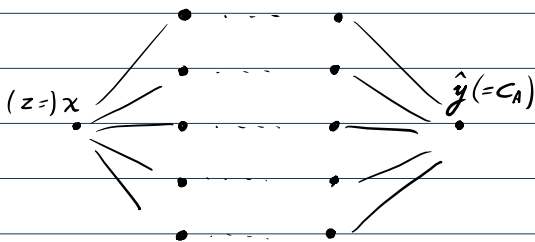
$$v_z \frac{\partial C_A}{\partial z} = r_A \Rightarrow v_z \frac{d}{dz} \left(\frac{F_A}{v_z A} \right) = r_A \Rightarrow \frac{dF_A}{dv} = r_A \quad \hat{C}_A = \frac{C_A}{C_{A_0}}, \quad \hat{z} = \frac{z}{L}$$

$$F_A = C_A Q = C_A v_z A \Rightarrow C_A = \frac{F_A}{v_z A}$$

$$\leadsto \frac{dF_A}{dv} = -k C_A^n \Rightarrow Q \frac{dC_A}{dv} = -k C_A^n \Rightarrow \frac{Q}{A} \frac{dC_A}{dz} = -k C_A^n \Rightarrow \frac{Q}{A} \frac{dC_A}{dz} + k C_A^n = 0 \Rightarrow \frac{Q}{A} \frac{C_{A_0}}{L} \frac{d\hat{C}_A}{d\hat{z}} + k C_{A_0}^n \hat{C}_A^n = 0$$

$$\Rightarrow \frac{Q C_{A_0}}{A L} \frac{1}{k C_{A_0}^{n-1}} \frac{d\hat{C}_A}{d\hat{z}} + \hat{C}_A^n = 0 \Rightarrow \frac{Q}{k C_{A_0}^{n-1} A L} \frac{d\hat{C}_A}{d\hat{z}} + \hat{C}_A^n = 0 \Rightarrow \hat{z} \rightarrow x \quad \boxed{\frac{1}{Da} \frac{dC_A}{dz} + C_A^n = 0}$$

$$\boxed{L_{PFR} = \left\| \frac{1}{Da} \frac{d\hat{y}}{d\hat{z}} + \hat{y}^n \right\|^2, \quad L_{data} = \frac{1}{N} \|y - \hat{y}\|^2, \quad L_{bc} = |\hat{y}[0] - y_0|^2, \quad L_{PINN} = L_{data} + \alpha L_{PFR} + \beta L_{bc}}$$



Training

Inputs: z coordinates ($1 \times N_p$), C_A values ($1 \times N_p$)

Find ground truth:

$$\frac{1}{Da} \frac{dC_A}{dz} + C_A^n = 0 \Rightarrow \frac{dC_A}{dz} = -Da C_A^n \Rightarrow \frac{dC_A}{C_A^n} = -Da dz \Rightarrow \int_1^{C_A} \frac{dC_A}{C_A^n} = -Da z$$

$$\text{For } n > 1: \quad \left[-\frac{1}{n-1} \frac{1}{C_A^{n-1}} \right]_1^{C_A} = -Da z \Rightarrow \frac{1}{n-1} \left(1 - \frac{1}{C_A^{n-1}} \right) = -Da z \Rightarrow 1 - \frac{1}{C_A^{n-1}} = -Da(n-1)z \Rightarrow$$

$$\frac{1}{C_A^{n-1}} = 1 + Da(n-1)z \Rightarrow \boxed{C_A(z) = \left[1 + Da(n-1)z \right]^{\frac{1}{1-n}}, \quad n > 1}$$

$$\text{For } n=1: \quad \int_1^{C_A} \frac{dC_A}{C_A} = -Da z \Rightarrow \ln C_A - \ln 1 = -Da z \Rightarrow \boxed{C_A(z) = e^{-Da z}, \quad n=1}$$

For $n=0$: $\int_1^{c_A} dc_A = -Da z \Rightarrow c_A - 1 = -Da z \Rightarrow c_A(z) = 1 - Da z, n=0$